A SEPARATED GROUP SPACE OVER K BECOMES A SCHEME OVER THE ALGEBRAIC CLOSURE

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Editorial note. No omission markers in-span; footnote records that decent (e.g. locally separated or quasi-separated) suffices via the separatedness lemma.

Allowed standard prerequisites (editorial).

- Schematic locus of an algebraic space. Properties of Spaces, lemma subscheme and proposition locally-quasi-separated-open-dense-scheme (cross-chapter, named). G_K' is dense open, in particular nonempty after base change.
- Jacobson rational points after large transcendental extension. Varieties, lemma make-Jacobson (cross-chapter, named). Produces K-rational points on U_K.

Source reference. Original lemma 047L: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 909–909. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B84: focused-seeds/geometry/S9-smg-group-space-scheme-over-kbar/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JH: focused-seeds/geometry/S9-smg-group-space-scheme-over-kbar/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NH: focused-seeds/geometry/S9-smg-group-space-scheme-over-kbar/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0479: focused-seeds/geometry/S9-smg-group-space-scheme-over-kbar/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field with algebraic closure $\overline{k}$. Let G be a group algebraic space over k which is separated*[1]*. Then $G_{\overline{k}}$ is a scheme.*

Proof. By Spaces over Fields, Lemma 0B84 it suffices to show that G_K is a scheme for some field extension K/k. Denote $G'_K \subset G_K$ the schematic locus of G_K as in Properties of Spaces, Lemma 03JH. By Properties of Spaces, Proposition 06NH we see that $G'_K \subset G_K$ is dense open, in particular not empty. Choose a scheme U and a surjective étale morphism $U \to G$. By Varieties, Lemma 0479 if K is an algebraically closed field of large enough transcendence degree, then U_K is a Jacobson scheme and every closed point of U_K is K-rational. Hence G'_K has a K-rational point and it suffices to show that every K-rational point of G_K is in G'_K. If $g \in G_K(K)$ is a K-rational point and $g' \in G'_K(K)$ a K-rational point in the schematic locus, then we see that g is in the image of G'_K under the automorphism

$$G_K \longrightarrow G_K, \quad h \longmapsto g(g')^{-1}h$$

of G_K. Since automorphisms of G_K as an algebraic space preserve G'_K, we conclude that $g \in G'_K$ as desired. □

[1]It is enough to assume G is decent, e.g., locally separated or quasi-separated by Lemma 047L.